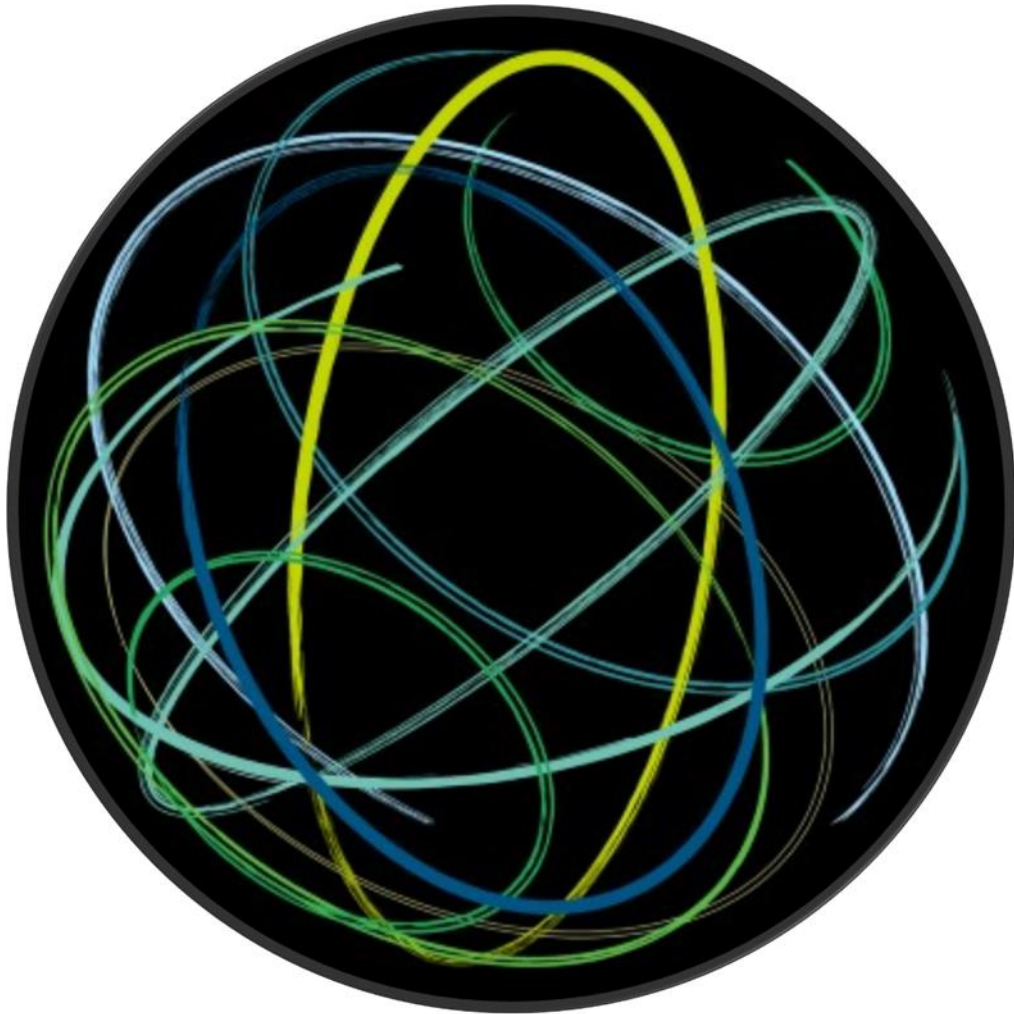




Flight Reservation System

Version Dated: 12.2025

Table of Contents

Problem Statement	3
Project Overview	3
Objectives	3
Tools and Technologies	3
Architecture and Flow Diagram	3
Database Structure	4
Functional Requirements	4
User Management	4
Admin Module	4
User Module	4
Wireframes	5
Home Page	5
Search Flight	5
Sign Up Page	6
Login Page	6
Book Flight – Booking Form	6
Book Flight – Add Passenger	7
Book Flight – Booking Confirmation	7
Admin Dashboard	8
Add Flight page	8
View Flights	8
User Dashboard	9
My Bookings page	9
View Tickets	9
Non-Functional Requirements	9
Guidelines	10
Milestones and Evaluation Criteria	10

Problem Statement

The complexity and demand for accurate, real-time air travel booking highlight the need for a robust flight reservation system. This project case study focuses on developing a web application to streamline the booking, ticketing, scheduling, and management processes for flights, supporting efficiency and an enhanced user experience.

Project Overview

This project involves developing a Spring-based Flight Reservation System. It enables users to browse, book, and cancel flight tickets, while administrators can add and manage flights and bookings. The system leverages Spring MVC, MySQL, and follows best practices for modular and secure application design.

Objectives

- Apply and demonstrate key concepts of Spring MVC and JDBC.
- Enable secure user and admin workflows (booking, management)
- Practice real-world database schema design, exception handling, and unit testing

Tools and Technologies

- Framework: Spring (MVC), Java
- UI: Spring Forms with JSP
- Database: MySQL
- ORM/Integration: Spring JDBC
- Build Tool: Maven
- IDE: Eclipse/STS/IntelliJ

Architecture and Flow Diagram

- **Presentation Layer:** JSPs powered by Spring MVC controllers.
- **Business Logic:** Spring services for flight, booking, and user modules.
- **Data Layer:** MySQL accessed via Spring JDBC.

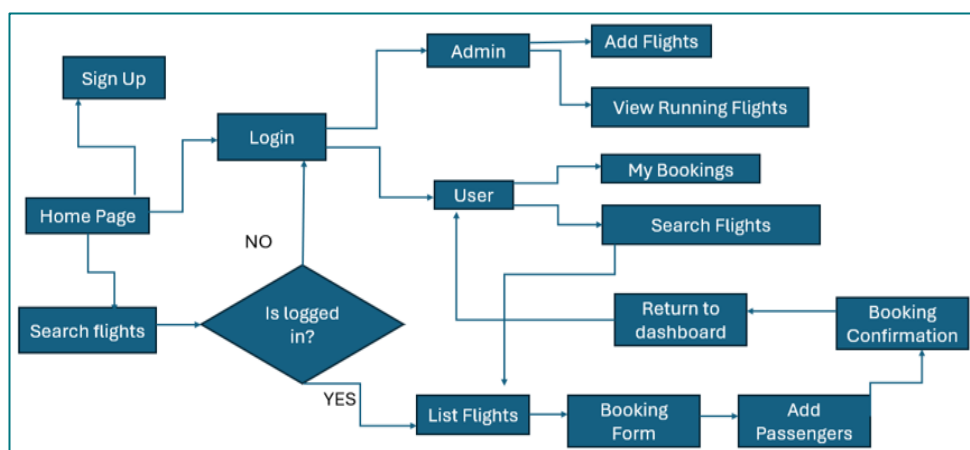
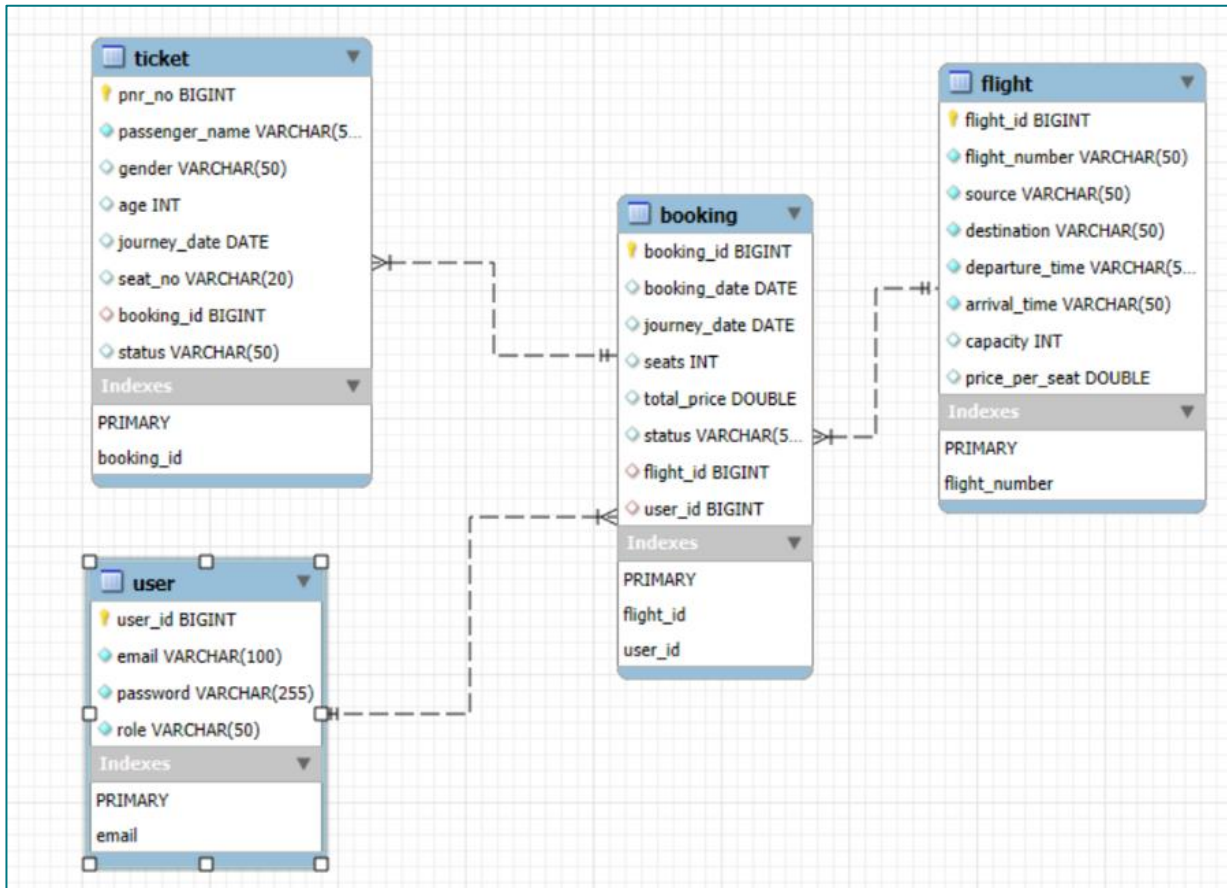


Figure 1: Flight Reservation System

Database Structure

- **Users Table:** Stores user accounts and roles.
- **Flights Table:** Manages flight info, source/destination, seat capacity, prices.
- **Bookings Table:** Tracks each booking transaction.
- **Tickets Table:** Holds data for each passenger's ticket.



Functional Requirements

User Management

- **Signup:** User registers with a unique email, password, and role (User/Admin).
- **Login:** Secure access for users and admins.

Admin Module

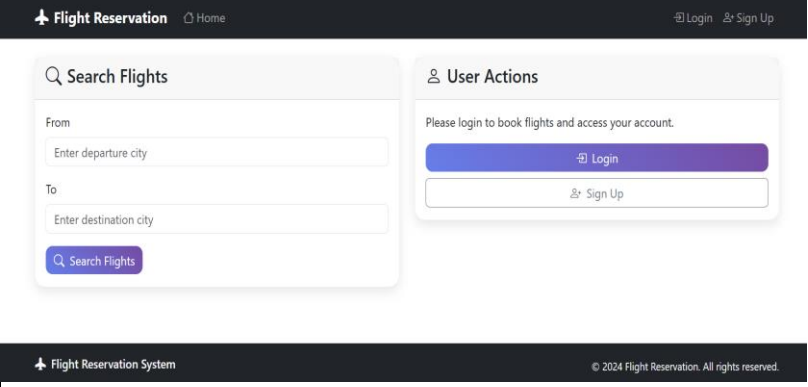
- Add new flights (all flights run daily, seat capacity: 20 by default).
- View all flights scheduled (Upcoming as well as Completed Trips).

User Module

- Search for flights by source and destination.
- Book flights: Generate individual tickets for each passenger.
- Cancel bookings or specific tickets (which must update available seat counts).
- View ticket details

Wireframes

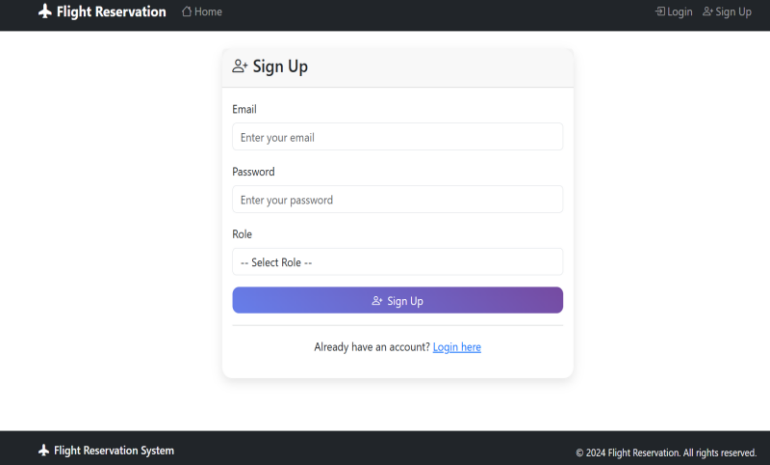
Home Page

Screen	Description
 The Home Page wireframe features a dark header with 'Flight Reservation' and 'Home' links, and 'Login' and 'Sign Up' buttons. The main content area is divided into two columns. The left column contains a 'Search Flights' form with 'From' and 'To' input fields, each with a placeholder 'Enter departure city' and 'Enter destination city' respectively, and a 'Search Flights' button. The right column contains a 'User Actions' section with the text 'Please login to book flights and access your account.' and two buttons: 'Login' and 'Sign Up'. The footer includes 'Flight Reservation System' and '© 2024 Flight Reservation. All rights reserved.'	• This will be the landing page of the application. • Search Flights form should be validated using Spring Forms. • Signup & login links should navigate to their respective page.

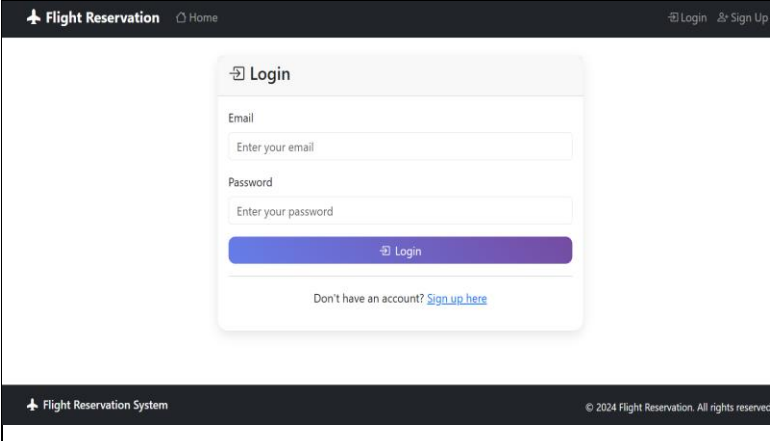
Search Flight

Screen	Description																				
✈️ Flight Reservation 🏠 Home 👤 Login 👤 Sign Up 🔍 Search Flights From Enter departure city To Enter destination city 🔍 Search Flights 👤 User Actions Please login to book flights and access your account. 👤 Login 👤 Sign Up ✈️ Available Flights <table><tr><th>Flight</th><th>Route</th><th>Schedule</th><th>Price</th><th>Action</th></tr><tr><td>AI101</td><td>Bangalore → Hyderabad</td><td>Dep: 19:49 Arr: 20:49</td><td>\$5000.0</td><td> ✈️ Book </td></tr><tr><td>IN201</td><td>Bangalore → Hyderabad</td><td>Dep: 18:41 Arr: 20:41</td><td>\$5000.0</td><td> ✈️ Book </td></tr><tr><td>AI304</td><td>Bangalore → Hyderabad</td><td>Dep: 11:25 Arr: 11:27</td><td>\$5600.0</td><td> ✈️ Book </td></tr></table> ✈️ Flight Reservation System © 2024 Flight Reservation. All rights reserved.	Flight	Route	Schedule	Price	Action	AI101	Bangalore → Hyderabad	Dep: 19:49 Arr: 20:49	\$5000.0	✈️ Book	IN201	Bangalore → Hyderabad	Dep: 18:41 Arr: 20:41	\$5000.0	✈️ Book	AI304	Bangalore → Hyderabad	Dep: 11:25 Arr: 11:27	\$5600.0	✈️ Book	• On clicking the Search Flights button, it should list all the available flights for the entered Source and Destination. • On clicking the Book button, it should redirect to the Booking Form. If the user is not logged in, the system should redirect to the login page. • On successful login only, it should redirect to the Booking Form Page.
Flight	Route	Schedule	Price	Action																	
AI101	Bangalore → Hyderabad	Dep: 19:49 Arr: 20:49	\$5000.0	✈️ Book																	
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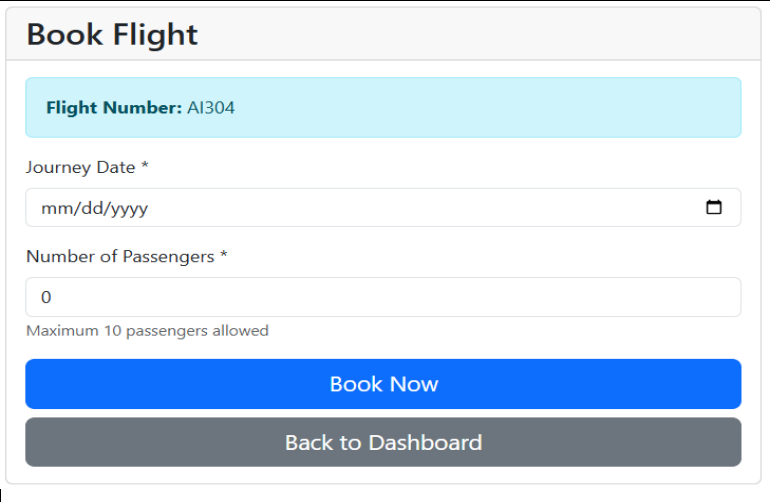
Sign Up Page

Screen	Description
	- Email and Password should be entered in proper format. - All three fields are required and must be validated using Spring Forms. - Role field should be selected as “Admin” or “User”. - Validation messages should be shown when the user interacts with the form fields.

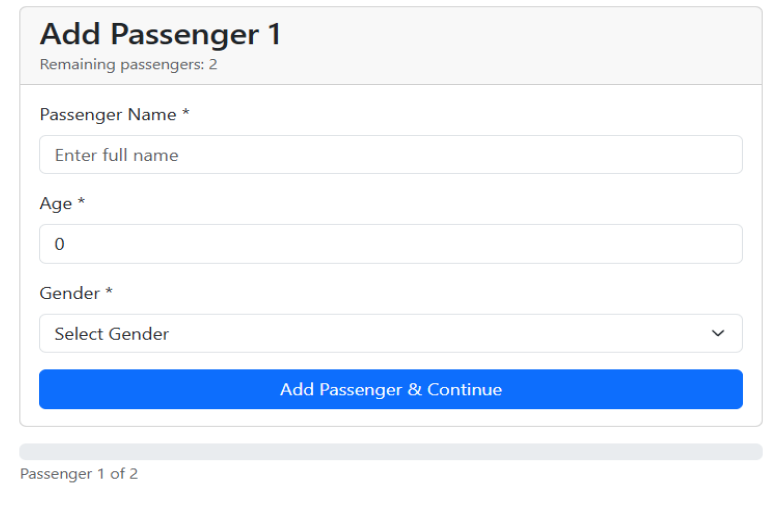
Login Page

Screen	Description
	- Email and Password should be entered in proper format. - All fields are required and must be validated using Spring Forms. - Validation messages should be shown when the user interacts with the form fields.

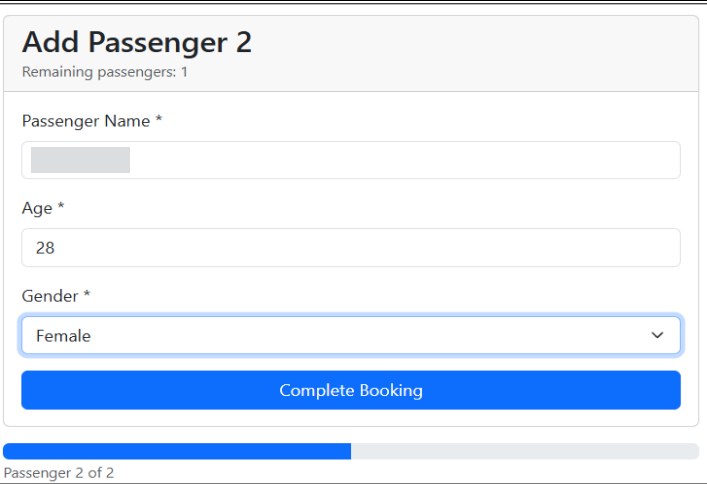
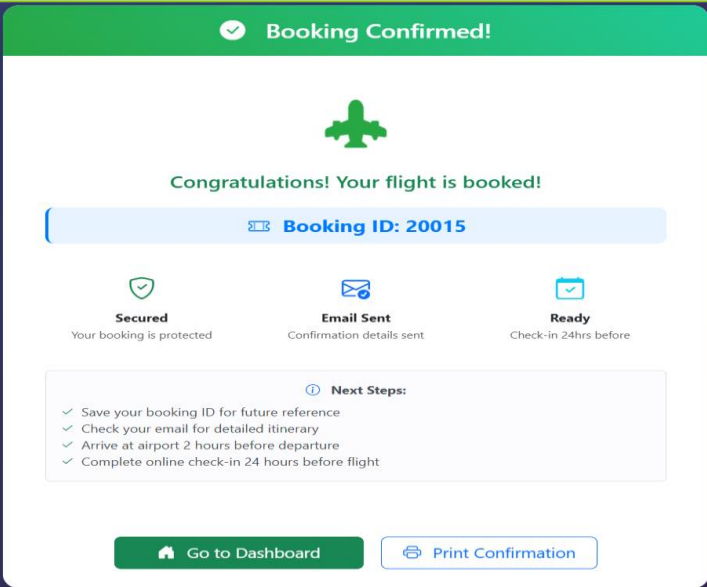
Book Flight – Booking Form

Screen	Description
	- All fields in this form are mandatory and must be validated using Spring Forms. - After selecting Journey date, Number of passengers, and clicking on Book Now button, check for the availability of seats. - If available seats < # passengers, then booking should not be allowed and an error message should be displayed on the screen or else it should redirect to Add Passenger Page.

Book Flight – Add Passenger

Screen	Description
	- All fields in this form are required and must be validated using Spring Forms. - If more than one passenger is there, then that many Add Passenger screens should appear.

Book Flight – Booking Confirmation

Screen	Description
	- All fields in this form are required and must be validated using spring forms. - In this last passenger screen, on clicking Complete Booking button, a booking confirmation page should appear as shown in the next screen.
	

Admin Dashboard

Screen	Description
	- Admin dashboard should have options to “Add New Flight” and “View Flights”. “Add flights” link should redirect to “Add Flight” form. “View flights” link should redirect to “View Flights” page.

Add Flight page

Screen	Description
	- All the fields of “Add Flight” form are required and must be validated using spring form. - Flight Number field should be unique.

View Flights

Screen	Description
	- View flights page should have two tabs: - Upcoming Journeys - Completed Journey - Only flights for which bookings are done should be displayed here. - All the details should be shown as in the screen displayed.

User Dashboard

Screen	Description
	- User dashboard should have two links: “My Bookings” and “Search Flights” “My bookings” link should redirect to “My Bookings” page. - Search Flights link should redirect to Home Page - “Search Flights page” (without displaying the login/signup/User Actions)

My Bookings page

Screen	Description
	- My Bookings page should list all the bookings made by the logged-in user only. - It should have two functions: - View Ticket (refer 10.7.2) - Cancel Ticket – - Confirmed seats must be released and - Cancel button must be disabled upon successful cancellation

View Tickets

Screen	Description
	Number of tickets generated should be based on the number of passengers mentioned in a particular booking.

Non-Functional Requirements

- Security:** Secure Authentication and data protection
- Performance:** Fast and responsive user experience
- Scalability:** Designed for future enhancements

Best Practices

- Dependency Injection for loose coupling
- Centralized exception handling (ControllerAdvice)
- Logging and Form validation
- Unit testing for Sign Up and Login functionalities
- Create reusable components
- Maintain consistent Angular coding styles and naming conventions
- Add comments wherever needed

Guidelines

- Make use of best coding practices.
- Keep UI simple and user-friendly.
- You can make assumptions and incorporate additional features or functionalities.
- A simple but working prototype is preferred over a nonworking but good-looking application.
- The screenshots provided in this document are only to enable your understanding and can be used as a reference to think through the logic and develop the application.
- Add relevant comments for proper documentation.
- Follow the naming conventions.
- Build reusable components.

Milestones and Evaluation Criteria

Milestone#	Milestone Description
Milestone 1	• Home Page functionality • Search and List Flights functionalities
Milestone 2	Login and Sign-Up functionalities
Milestone 3	Book Flight Functionality: • Booking Form • Add Passenger • Booking Confirmation
Milestone 4	Admin Dashboard: • Add Flights • View running flights
Milestone 5	User Dashboard: • My Bookings • View Tickets
Milestone 6	Implement Global Exception Handling

Milestone 7	Implement Unit Testing for Login and Sign-Up Screens
Milestone 8	Push the build to your GitHub repository
Technical Interview/Viva	Technical Concepts, Implementation, Discussion of Assumptions, Functionalities, and Validations



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